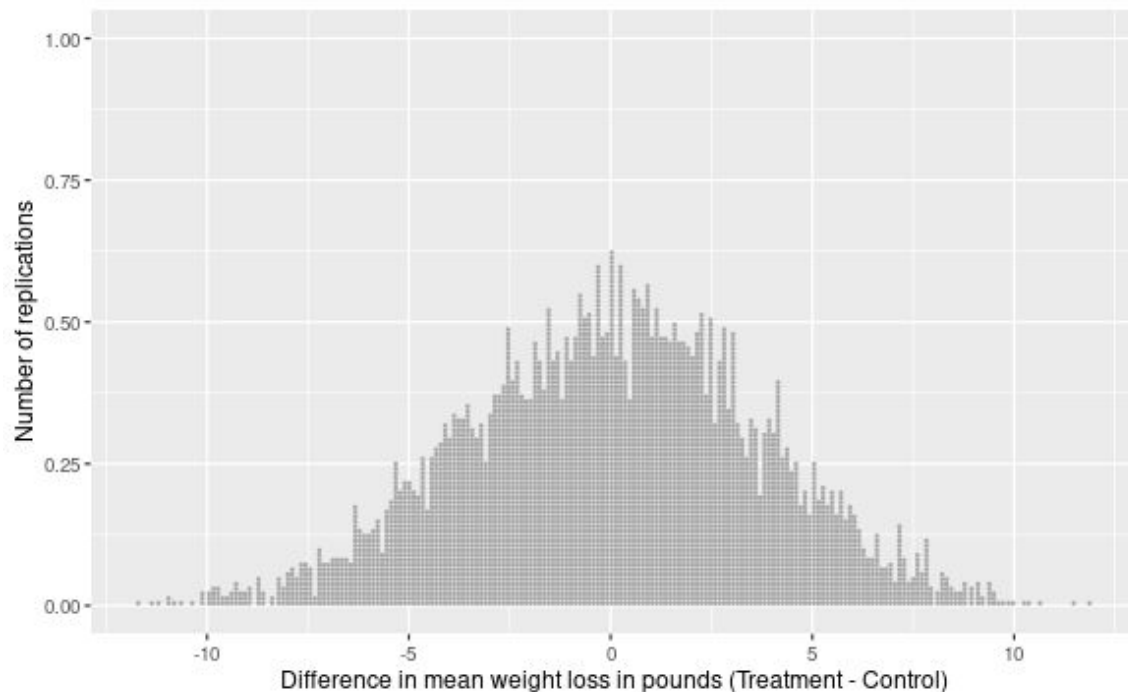


2-Weight Loss Study

MATH 213

Group 1

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?

5000

What does each dot represent?

Difference in mean incentive when assuming the null hypothesis is true

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

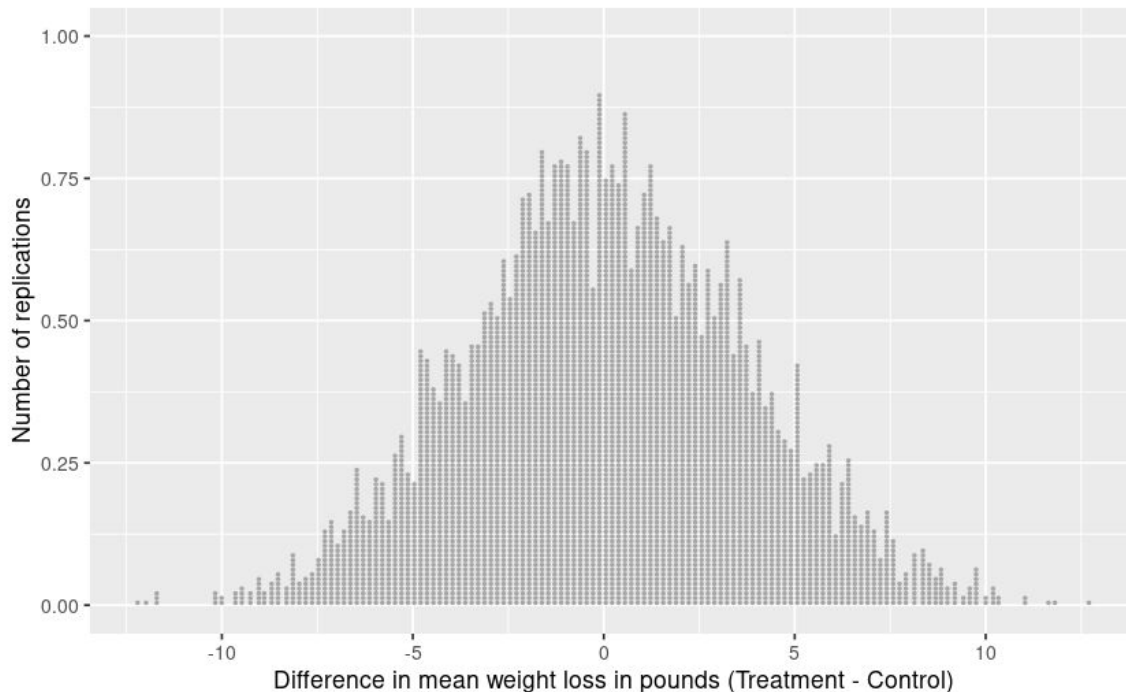
Almost impossible

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

After seeing the results of 5000 simulations, it seems very plausible. Just looking at the graph and thinking about a bell curve, to get 11.755 pounds would be under 0.1%

Group 2

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?

5000

What does each dot represent?

Difference in mean incentive for a particular hypothetical replication under the assumptions that the treatment has no effect on a participant's weight loss at 16 weeks.

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

Not very common. Based on our R calculation, there were only 2 cases that were greater than 11.755 pounds.

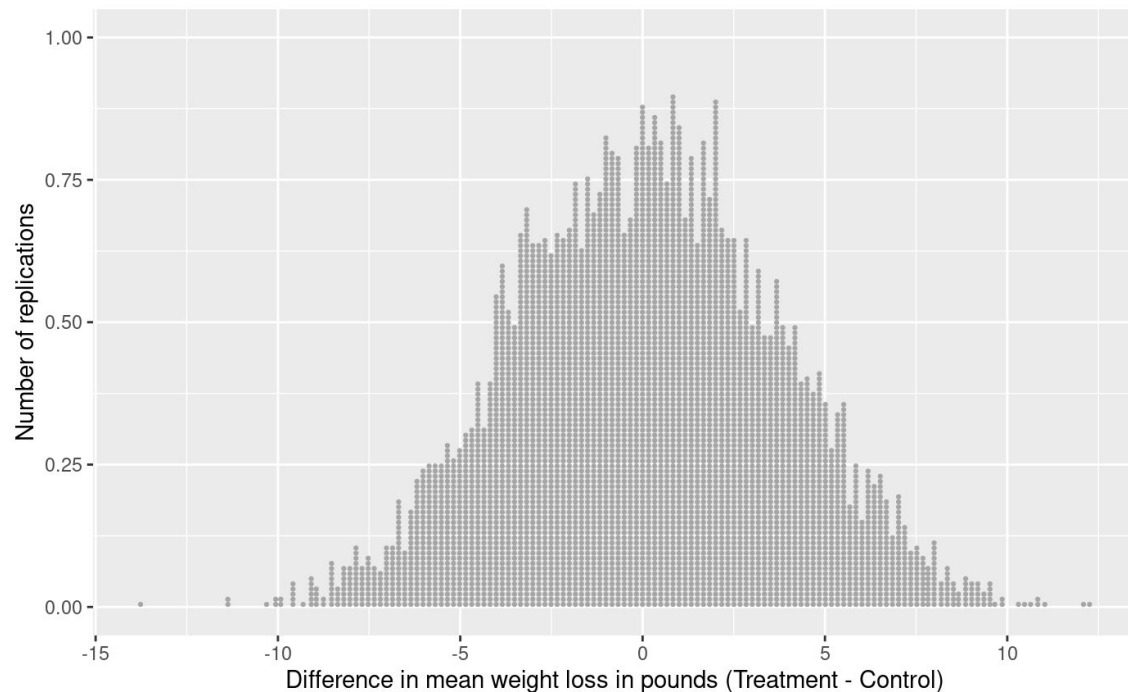
Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

The null hypothesis does not seem plausible in this case, since only about 2 cases out of 5000 simulations under the null hypothesis were able to replicate the results of our experiment.

Group 3

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.

How many dots are on this dotplot?
5000



What does each dot represent?
A difference in one simulation between the average treatment and the average control weight.

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

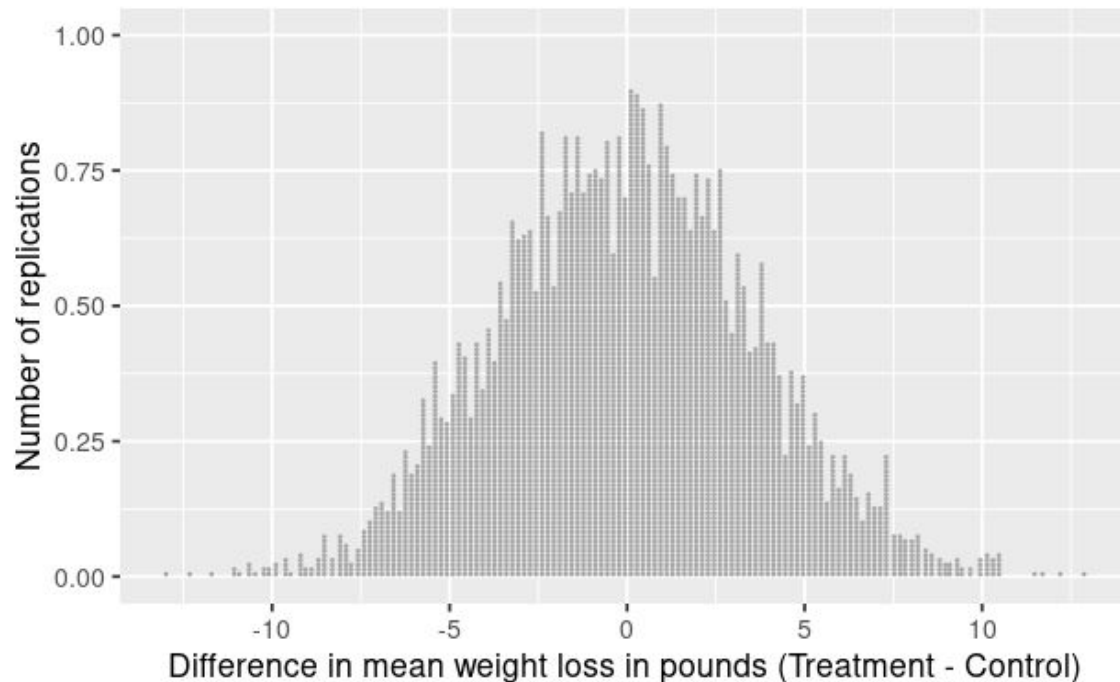
There was a 0.006 percent chance

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

No, in the simulation there was a less than 1 percent chance that the outcome occurred over 5000 simulations. Hinting towards a great possibility that the study did show incentive weight loss had an effect on weight loss.

Group 4

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?
5000

What does each dot represent?
Each dot is the difference of the means between the control and incentive group for a single simulated experiment

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

~~4/5000 = 1/1250~~

~~About 1 in every 1250 experiments; .08% of experiments~~

2/5000 = 1/2500

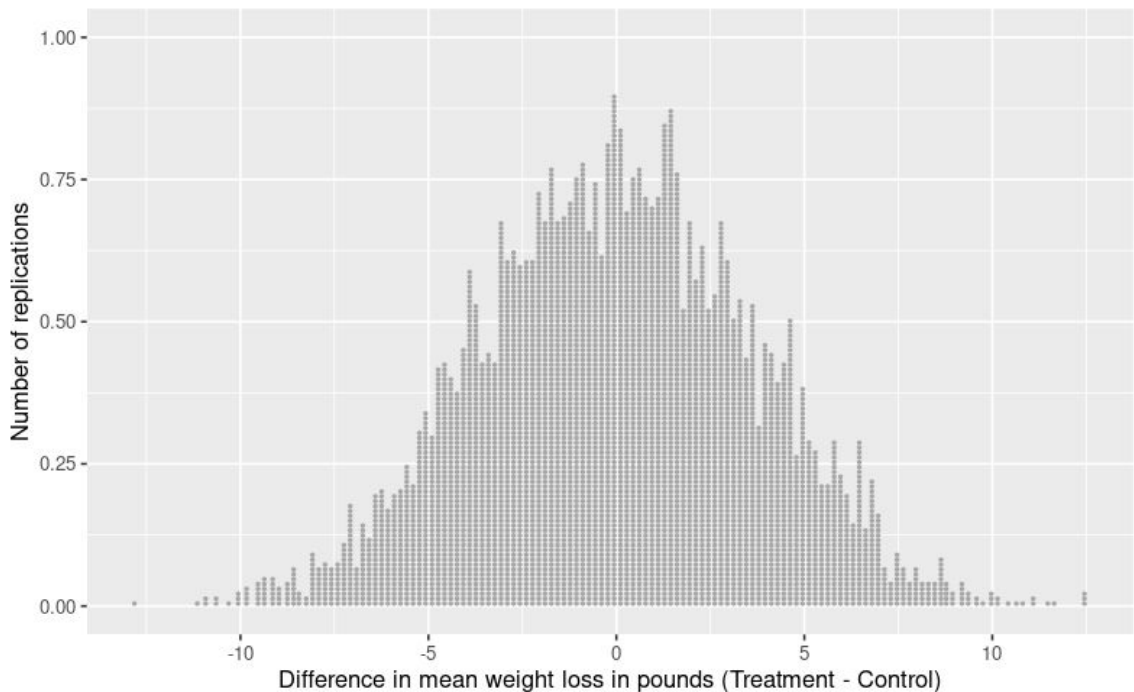
About 1 in every 2500 experiments; .04%

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

No, because under this assumption that the group has no effect on weight loss, only about 1 in every 2500 experiments have a similar difference in means.

Group 5

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?

5000

What does each dot represent?

Each dot represents a randomized simulation of the difference in mean weight loss in pounds of Treatment vs. Control when assuming the null hypothesis is true, with each sample having 36 cases

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

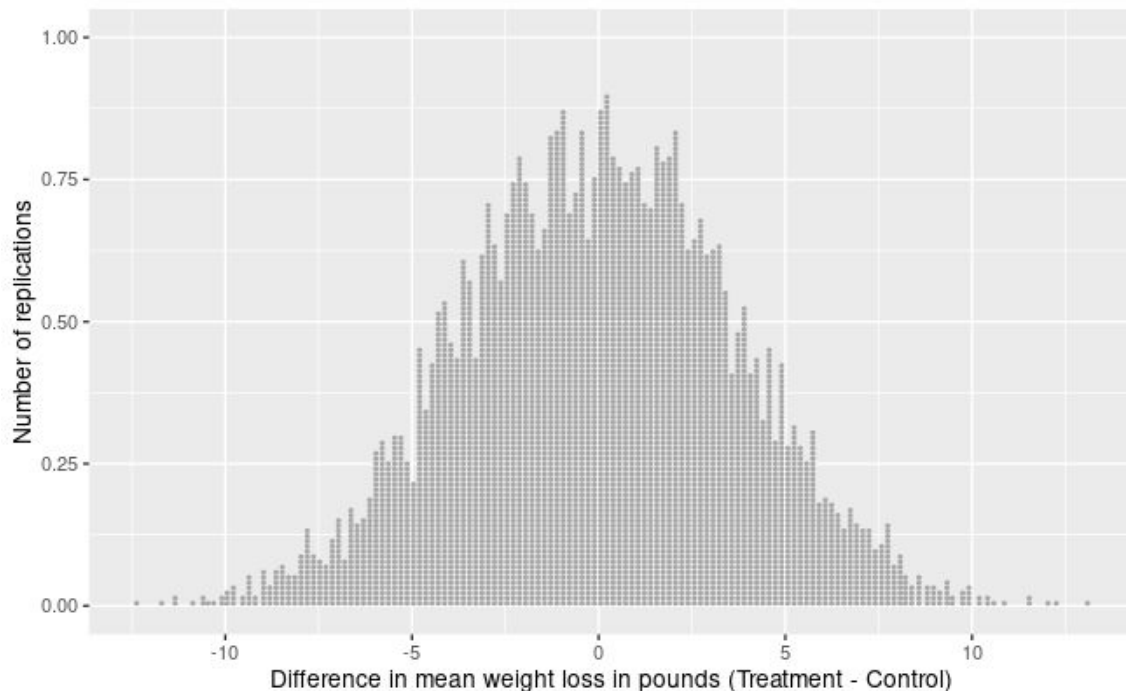
3/5000 = 0.0006, 0.06% chance of a replication having a difference in means that was as or more extreme than our actual value

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

No, since the chance of replication is extremely low the hypothesis is not plausible.

Group 6

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?

5,000

What does each dot represent?

The performance difference between the incentive and control groups in the 5000 simulations assuming that the null hypothesis is held to be true.

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

There were only 3 replications that had a greater or equal difference in means than the actual value of 11.755.

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

No, because the replicated data shows that only 0.06% of simulations had a similar difference in mean than the actual value.

```
```{r}
many_reps %>%
 filter(diff >= 11.755) %>%
 select(replicate, diff)
```
```

A tibble: 3 × 2

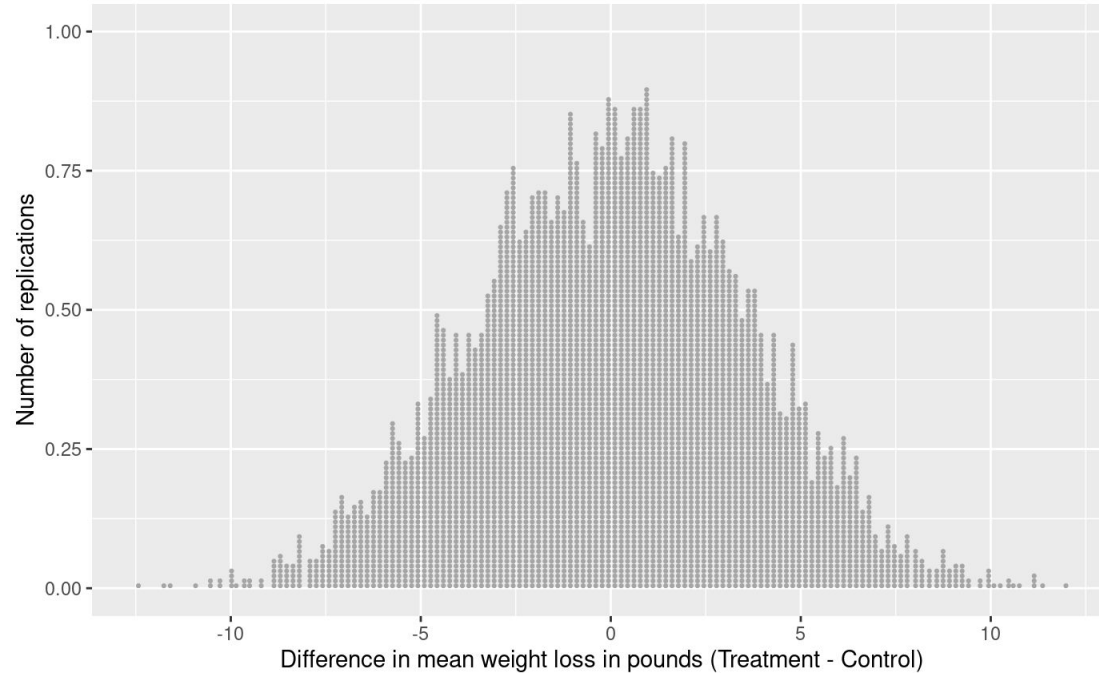
Groups: replicate [3]

| replicate
<int> | diff
<dbl> |
|--------------------|---------------|
| 199 | 11.81115 |
| 1175 | 13.42724 |
| 4957 | 11.75542 |

3 rows

Group 7

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?

5000

What does each dot represent?
Each dot represents the simulated result of the amount of weight lost in the treatment - the amount of weight lost in the control under the assumption that there is no difference in groups

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

Near 0. Out of 5000 replications, only 1 was above 11.755.

4050 11.97833

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

Group 8

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.

How many dots are on this dotplot?

What does each dot represent?

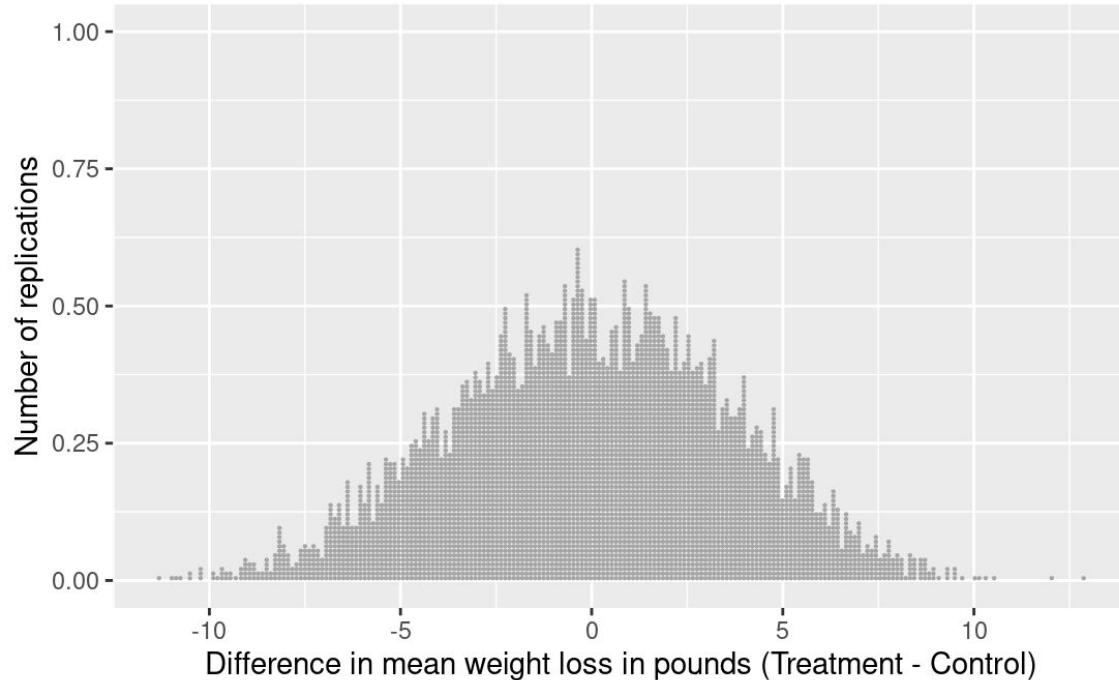
How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

Group 9

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.



How many dots are on this dotplot?
5000

What does each dot represent?
Each dot represents the difference in incentive-control per each replication assuming the null hypothesis is true

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

From what the graph presented, very nearly impossible

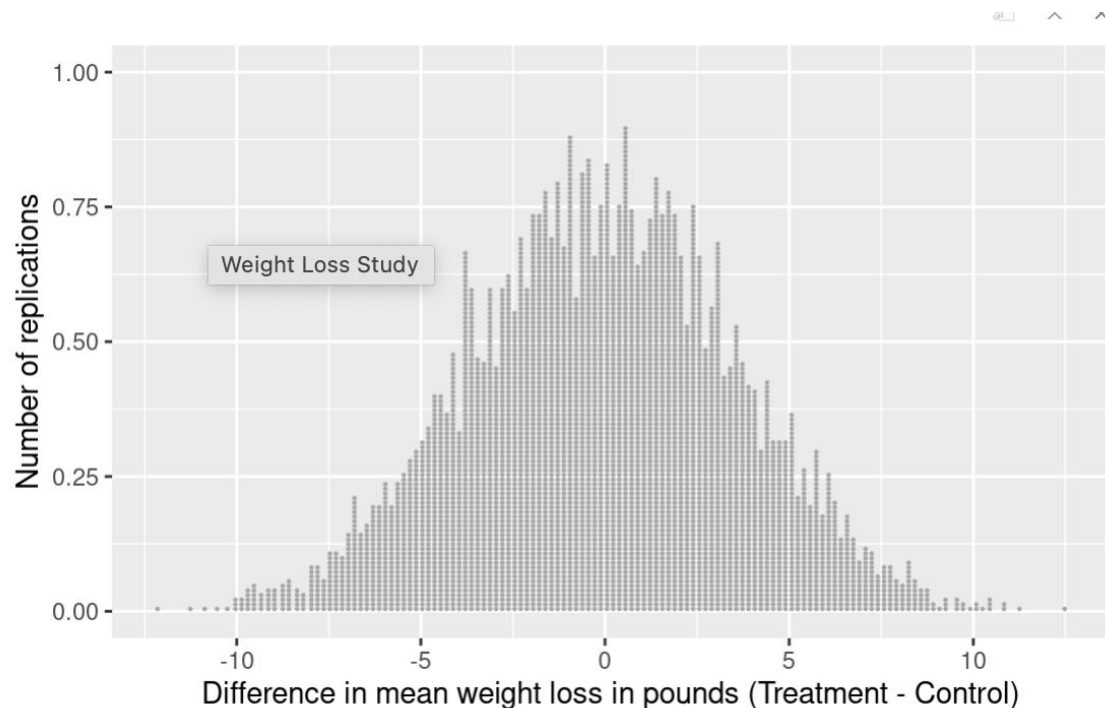
Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

No because if it was impossible then there should have been far more likely of a chance that the hypothesis mean would show up. Yet for the null effect it was very nearly impossible, suggesting special circumstances needed to be involved in order to produce said number

Group 10

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.

How many dots are on this dotplot?
5000



What does each dot represent?
Each dot represents The difference in mean weight loss of treatment vs control.

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

Group 11

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.

How many dots are on this dotplot?

108 Dots

What does each dot represent?

Each dot represents the value of “difference in incentive and control” from several artificial replications of the experiment and plotted.

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

The group that they were placed in had to have some effect as the extreme value of 11.755 pounds is nowhere to be seen on our dot plot representation after running 108 times. P-value is equal to $0/108 = 0$. Meaning it is entirely improbable for such a high result.

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.

Group 12

Copy and paste here the dotplot showing many hypothetical replications of the weight-loss experiment under the hypothesis that the treatment had no effect on the weight loss of any subject at 16 weeks.

How many dots are on this dotplot?

What does each dot represent?

How common would it be to see an experimental result like the actual experiment under the hypothesis?

(Hint: compute the number of replications where the difference in means was as or more extreme than the actual value of 11.755 pounds. Perform a calculation in R and use numbers to answer!)

Given the actual experimental data, does the hypothesis that the incentive weight loss plan had “no effect” seem plausible? Explain your thinking.